

# Aaron Mohammed

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https://github.com/aaronmohammed

## SKILLS

### Languages

Python, Go, Java, SQL, Bash

### Tech

Python, Go, Java, PostgreSQL, Redis, MongoDB, GraphQL, REST APIs

### Tools

Docker, Kubernetes, AWS (EC2, RDS, Lambda, S3), Git, Terraform, CircleCI, DataDog

## ACTIVITIES

### ByteScale

#### Community

Technical Lead  
2022 - Present

#### Open Source

#### Contributions

Active Contributor  
2021 - Present

## EDUCATION

**University of California, Berkeley** — *B.S. in Computer Science* 2017 - 2021

- Completed coursework in Distributed Systems, Database Systems, and Advanced Algorithms with focus on large-scale system design patterns

## EXPERIENCE

**Stripe** — *Backend Engineer II* Mar 2023 - Present

- Architected and deployed a distributed transaction reconciliation service in Go, reducing payment settlement discrepancies by 98.7% and recovering \$2.3M in disputed charges annually
- Optimized database queries for the billing pipeline by implementing strategic indexing and query caching, reducing P99 latency from 840ms to 120ms and handling 4.2x throughput
- Led migration of critical payment routing microservice from monolithic Node.js to containerized Go service on Kubernetes, improving deployment velocity by 65% and reducing memory consumption by 52%

**Notion** — *Backend Engineer* Jul 2021 - Feb 2023

- Built and maintained 8 core APIs serving 15+ million monthly active users, achieving 99.94% uptime SLA and handling peak throughput of 85,000 requests per second
- Designed real-time collaboration backend using event sourcing and CRDT principles, enabling simultaneous multi-user editing with <200ms sync latency across 3 geographic regions
- Implemented comprehensive rate limiting and quota management system using Redis, protecting infrastructure from abuse and reducing DDoS impact by 73%

**Figma** — *Junior Backend Engineer* Jan 2021 - Jun 2021

- Developed asynchronous file processing pipeline using Python and Celery, decreasing average export time from 8.4 seconds to 2.1 seconds for 50,000+ daily export operations
- Implemented caching layer using Redis for frequently accessed design component metadata, improving API response times by 61% and reducing database load by 44%
- Contributed to PostgreSQL schema optimization and implemented data partitioning strategy for analytics tables, improving query performance by 3.8x for historical reports

## PROJECTS

**Distributed Trace Aggregation System** — *Go, Kafka, OpenTelemetry, Protocol Buffers, BigQuery* Aug 2023 - Oct 2023

- Engineered high-throughput trace ingestion service processing 2.4 billion spans daily with <500ms end-to-end latency, enabling real-time performance monitoring across 120+ microservices
- Implemented intelligent trace sampling algorithm reducing storage costs by 42% while maintaining statistical accuracy for anomaly detection and SLA tracking

**Database Connection Pool Manager** — *Java, PostgreSQL, Spring Boot, Micrometer* Nov 2022 - Dec 2022

- Created adaptive connection pooling library with predictive scaling, reducing connection exhaustion incidents by 89% and improving peak throughput from 12,500 to 18,700 queries per second
- Implemented comprehensive observability with detailed connection lifecycle metrics, reducing MTTR for database issues by 56%

**Event Sourcing Framework** — *Python, PostgreSQL, Apache Kafka, Pydantic* Mar 2022 - May 2022

- Developed open-source event sourcing framework adopted by 340+ developers, simplifying event-driven architecture implementation with automatic event versioning and replay capabilities
- Reduced implementation time for event-driven systems from 3 weeks to 4 days for new backend engineers at Notion